

F24-087-D-Dex'sLab

Dexter's Laboratory: A VR Laboratory Simulator

Project Team

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Chapter 1

Introduction

This proposal paper describes the goals, salient characteristics, potential application, and driving force behind Dexter's Laboratory (Dex's Lab), a 3D virtual chemistry lab intended for O-level and 9–10 pupils. By offering an immersive virtual reality (VR) environment where students may safely experiment with chemical ideas in a simulated, real-life laboratory setting prior to engaging in actual laboratories, the platform seeks to improve students' learning experiences. By describing the program's technological design and contrasting its intellectual advantages with those of conventional learning settings, this document aims to demonstrate why such an application is necessary.

Key topics in chemistry are sometimes difficult for students to understand in traditional classroom settings because of things like a lack of lab equipment, short learning sessions, safety hazards, and insufficient direction while handling chemicals and equipment. In order to overcome these obstacles, Dex's Lab provides a completely realistic virtual lab that replicates actual laboratory experiences in a secure setting. In order to gain confidence using lab equipment, students can repeat experiments and learn from the practical outcomes while following detailed instructions and practicing on their own.

This document is primarily aimed towards various stakeholders. Starting with, this document intends to brief the **developers** with the basic understanding of technical and architectural aspects of the project. **Project Managers** would be responsible for reviewing the timelines, scope and the objectives of the project. This document also seeks to inform **users**, students and educators about the innovative advancements this application brings to the learning process. The **testers** ensure the performance, safety and availability of the application. Additionally, this article helps **documentation writers** produce technical documentation, end-user guides, and instruction manuals.

Dex's Lab, which uses technology to improve students' comprehension of chemistry,

marks a major breakthrough in educational innovation by bridging the gap between theoretical learning and practical application.

1.1 Problem Statement

We are developing Dexter's Laboratory: A VR laboratory Simulator to address the growing concerns about having fewer laboratories and resources available in many educational institutions, especially the ones that are facing budget constraints. Many students are driven by not getting actual hands-on experience/knowledge due to lack of funding, and improper or limited facilities which puts them in a limitation in life experience and how they apply this theoretical knowledge in practice. For this reason, our software addresses the uprising issue, allowing students to perform laboratory experiments in a virtual environment. Hence, the student is given the actual conditions introduced in the laboratory as an opportunity to conduct experiments that could not otherwise be carried out. In addition, it aligns with the Matric and O-level curriculum thus helping students to revise the concepts entertainingly. Furthermore, this system is convenient for schools and parents as they will be provided with an environment they require without having to restrict themselves from enacting physical labs.

1.2 Scope

The Dexter's Laboratory will be designed specifically for chemistry education in O Levels and Pakistan Matric standards, targeting students in classes 9-11. The scope of this project is confined to experiments and concepts within these curricula, setting clear boundaries for development and implementation.

The simulator will cover essential chemistry experiments and techniques, including:

- Cation and anion tests
- Acid-base titrations
- pH testing using various indicators
- Exothermic and endothermic reactions
- Proper handling and use of common laboratory apparatus for example pipettes, burettes, and thermometers

The project's scope is strictly limited to the chemistry curriculum for O Levels and Matric grades. It will not extend to other scientific disciplines or more advanced chemistry concepts beyond this educational level. The simulator will focus on providing

virtual hands-on experience for experiments that are part of the standard curriculum, ensuring relevance to the target student group.

In terms of language, the interface and instructions will be available in English since the curriculum and manuals are provided in English. The project scope includes creating accurate 3D models of laboratory equipment and chemicals relevant to the specified experiments, along with realistic simulations of chemical reactions, interactions, and processes.

The VR Lab Simulator will be developed exclusively for the Meta Quest platform, limiting its hardware compatibility but ensuring optimal performance on this specific device. While the system will be designed with the potential for future expansions, the current scope does not include experiments or concepts outside the defined curriculum.

Safety protocols and proper laboratory techniques will be integrated into the simulations, teaching students correct procedures without the risks associated with physical labs. However, the scope does not extend to replacing physical labs entirely but rather to supplementing traditional teaching methods and providing opportunities where physical labs are not feasible.

1.3 Modules

1.3.1 Module 1: Asset Creation

1. Laboratory Room/environment design
2. Equipment Asset Creation
3. Texture and Coloring of Assets
4. VR player movement
5. Virtual hands and basic object interaction

1.3.2 Module 2:Mechanic Design

1. Core chemical reaction simulation engine
2. Basic equipment handling (pipettes, burettes, thermometers)
3. Liquid pouring and measuring mechanics
4. Simple experiment procedure management

1.3.3 Module 3:Level Design

1. Acid-bases and Salts
2. pH testing with indicators
3. Cation and anion test implementation
4. Exothermic and endothermic reactions
5. Physical States of Matter
6. Chemical Reactivity
7. Organic Chemistry

1.3.4 Module 4: Application Design

1. Safety protocol system (virtual safety equipment, hazard warnings)
2. UI for instructions and feedback
3. Tutorial and guidance system
4. Main menu and experiment selection interface

1.3.5 Module 5: Testing

1. Validate the working of tools and equipment and smooth and measurable mechanics
to ensure accurate experiment execution.
2. Tests for smooth performance across platforms, ensuring frame rates are stable
and
no significant performance bottlenecks occur.
3. Ensure that the player movement within the VR environment is smooth, with

proper
collision detection.

4. Level-based testing to ensure the experiment sequence is integrated correctly

1.3.6 Module 6: Deployment and Optimization

1. Performance optimization
2. Graphics enhancement
3. Meta Quest integration using Meta Quest SDK
4. Final UI/UX refinements

1.4 User Classes and Characteristics

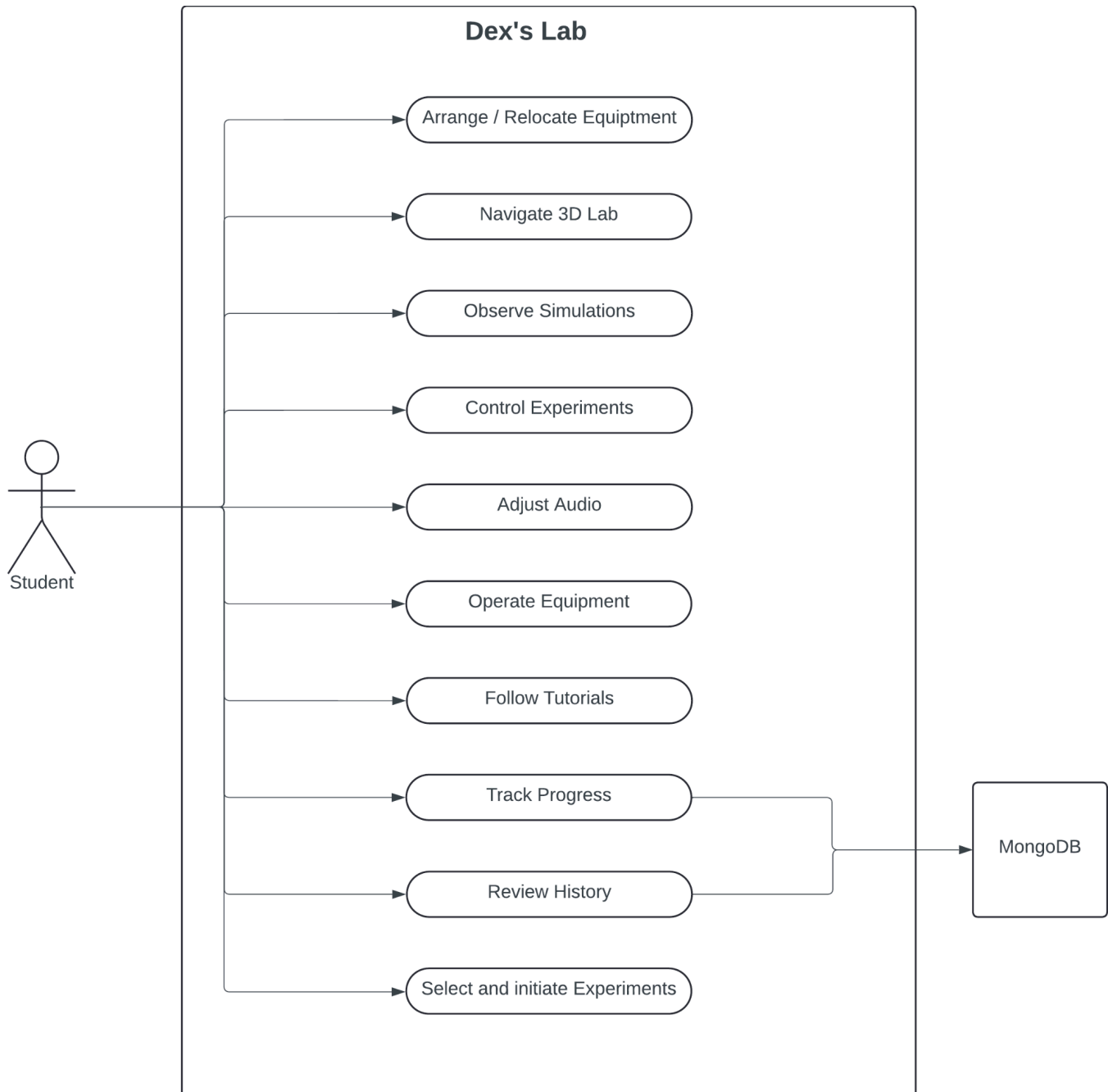
Students:

- The primary users who will interact with the virtual lab environment
- Engage in experiments, simulations, and learning activities
- Require guidance and support through tutorials, progress tracking, and access to pre-set experiments
- Need to be able to navigate the 3D lab space, operate equipment, and adjust settings
- Benefit from features like historical data review and self-directed learning opportunities

Chapter 2

Project Requirements

2.1 Use-case



2.1: Use-case diagram

2.1.1 Pick Up and Relocate Laboratory Equipment in the Virtual Lab

Description	As a Student, I want to be able to pick up and relocate laboratory equipment such as pipettes, burettes, and thermometers within the virtual lab environment, so that I can arrange the workspace to my preference and prepare for my experiments.
Actor:	Student
Goal:	The student wants to be able to pick up and relocate various laboratory equipment, such as pipettes, burettes, and thermometers, within the virtual lab environment.
Pre-conditions:	• The student has access to the Dexter's Laboratory: A VR Laboratory Simulator application. • The student must have started a lab session, through the experiments menu
Main Flow:	1. The student launches the Dexter's Laboratory application and joins the virtual lab activity by selecting one of the experiments in the experiments menu 2. The student is taken to the virtual laboratory environment, where they can see the various laboratory equipment on the lab benches and tables 3. Once inside, the student finds a fully-equipped lab and uses VR controllers to pick up and move equipment like pipettes, burettes, and thermometers around the workspace 4. The student can rearrange the equipment to set up their ideal virtual workstation, placing items wherever they like and adjusting their orientation 5. When done with a tool, the student simply puts it back down on the lab surface, ready to interact with other virtual lab equipment
Alternative Flows	If the student wants to reset the virtual lab to its default layout, they can use the options in the application menu

2.1.2 Move around in the 3d space of the virtual laboratory

Description	As a student, I want to freely move around the 3D virtual lab environment, so I can explore the space and access the various equipment to perform my experiments.
Actor:	Student
Goal:	The student wants to be able to freely move and navigate within the

	3D space of the virtual laboratory environment.
Pre-conditions:	• The student has access to the Dexter's Laboratory: A VR Laboratory Simulator application. • The student must have started a lab session, through the experiments menu
Main Flow:	1. The student launches the Dexter's Laboratory application and joins the virtual lab activity by selecting one of the experiments in the experiments menu 2. The student is transported to the 3D virtual laboratory environment, where they can see the lab space, including various equipment, workstations, and other features 3. The student uses the VR controllers to navigate and freely move around the virtual lab space 4. The student can walk around the lab, such as the chemical storage area, workbenches, or demonstration stations. 5. The student can look around the environment, to track their movement or focus on other things in the experiment

2.1.3 Conduct, Observe and Learn from the Simulated Chemical Reactions and Processes

Description	As a student, I want the ability to perform, observe, and learn from simulated chemical reactions and processes in the virtual lab, so I can develop a deeper understanding of chemistry concepts and principles.
Actor:	Student
Goal:	The student wants to conduct experiments, observe the simulated chemical reactions and processes, and learn from the virtual laboratory environment.
Pre-conditions:	• The student has access to the Dexter's Laboratory: A VR Laboratory Simulator application. • The student must have started a lab session, through the experiments menu
Main Flow:	1. The student launches the Dexter's Laboratory application and joins the virtual lab activity by selecting one of the experiments in the experiments menu 2. The student is then transported to the virtual laboratory environment and selects an experiment to conduct. 3. The student interacts with the virtual laboratory equipment, such as pipettes, burettes, and thermometers, to set up and

	perform the experiment. - The student can repeat the experiment or explore other pre-set demonstrations to expand the student's knowledge. - The student can also access additional learning resources, such as tutorials and explanations, to deepen the student's understanding of the chemistry concepts.
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2.1.4 Pause, start, and stop experiments

Description	As a Student, I want to be able to pause, start, and stop experiments in the virtual lab environment, so that I can better control the flow of the experiments and focus on learning.
Actor:	Student
Goal:	The student wants to have the ability to pause, start, and stop experiments within the virtual lab environment.
Pre-conditions:	- The student has access to the Dexter's Laboratory: A VR Laboratory Simulator application. - The student must have started a lab session, through the experiments menu
Main Flow:	- The student launches the Dexter's Laboratory app and joins the virtual lab activity by choosing one of the experiments from the menu or using the provided access code. - Once in the virtual lab environment, the student selects an experiment to perform. - The student sets up the chosen experiment and starts the simulation. - During the experiment, the student can pause the simulation at any time using the controls in the virtual lab interface. - If the student needs to completely stop the experiment, they can use the stop button, which will end the simulation and allow the student to reset or move on to a different experiment. - The student can repeat this process of pausing, starting, and stopping experiments as needed to optimize their learning experience.
Alternative Flow	If the student wants to reset the experiment to its initial state, they can use the button in the experiment menu.

2.1.5 Adjust Sound Levels in the Virtual Lab

Description	As a Student, I want to be able to adjust the sound levels in the
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	virtual lab environment, so that I can customize the audio to my preferences and optimize my learning experience.
Actor:	Student
Goal:	The student wants to have the ability to adjust the sound levels within the virtual lab environment.
Pre-conditions:	• The student has access to the Dexter's Laboratory: A VR Laboratory Simulator application.
Main Flow:	1. The student launches the Dexter's Laboratory application and joins the virtual lab activity by selecting one of the experiments in the experiments menu or using the provided access code 2. The student is transported to the virtual laboratory environment and selects an experiment to perform. 3. Student can use the buttons on the meta quest headset to adjust the volume or long press the meta button on either controller to bring up the menu 4. Student can then select the audio level that suits them best

2.1.6 Operate and Interact with Equipment in the Virtual Lab

Description	As a Student, I want to be able to operate and interact with the various equipment and instruments in the virtual lab environment, so that I can perform experiments, collect data, and deepen my understanding of laboratory techniques.
Actor:	Student
Goal:	The student wants to have the ability to operate and interact with the equipment and instruments available in the virtual lab environment.
Pre-conditions:	• The student has access to the Dexter's Laboratory: A VR Laboratory Simulator application. • The student must have started a lab session, through the experiments menu
Main Flow:	1. The student launches the Dexter's Laboratory application and joins the virtual lab activity by selecting one of the experiments in the experiments menu or using the provided access code. 2. The student is transported to the virtual laboratory 3. The student approaches a specific piece of equipment, such as a pipette, burette, balance, or thermometer, and uses the

	VR controllers to interact with it. 4. Student can perform various operations on the equipment such as a. Turn on / off b. Adjust level such as (Gas, Heat, Liquid flow) etc c. Collect measurements 5. The student can repeat this process with multiple pieces of equipment, setting up their virtual lab workspace and performing experiments. 6. As the student interacts with the equipment, they can observe the simulated reactions, processes, and data outputs in real-time.
Alternative Flow	If the student wants to reset or recalibrate the equipment, they can use the button in the menu.

2.1.7 Access Step-by-Step Tutorials and Guidance for Experiments in the Virtual Lab

Description	As a Student, I want to be able to access step-by-step tutorials and guidance for the experiments in the virtual lab environment, so that I can learn the proper procedures and techniques to conduct the experiments effectively.
Actor:	Student
Goal:	The student wants to have access to step-by-step tutorials and guidance to help them perform experiments in the virtual lab.
Pre-conditions:	● The student has access to the Dexter's Laboratory: A VR Laboratory Simulator application. ● The student must have started a lab session, through the experiments menu
Main Flow:	1. The student launches the Dexter's Laboratory application and joins the virtual lab activity by selecting one of the experiments in the experiments menu or using the provided access code. 2. The student is transported to the virtual laboratory 3. The student locates the "Tutorials and Guidance" section, which may be accessible through a menu, button, or icon within the virtual lab interface. 4. The student browses the available tutorials and selects the one corresponding to the experiment they want to perform. 5. The tutorial provides step-by-step instructions and guidance on how to set up and conduct the experiment, including: a. Identifying the required equipment and materials

	b. Explaining the proper techniques for using the equipment c. Outlining the sequence of steps to follow during the experiment d. Highlighting important safety considerations e. Providing tips and best practices for successful experimentation 6. The student can pause the tutorial, rewind, or replay specific steps as needed to ensure they understand the process. 7. As the student performs the experiment, they can refer back to the tutorial for ongoing guidance and support. 8.
Alternative Flow	1. If the student wants to exit / restart the tutorial they can do so using the button

2.1.8 Monitor Current Progress in Ongoing Virtual Lab Experiments

Description	As a Student, I want to be able to monitor the current progress of the experiments I'm doing in the virtual lab environment so that I can track the status of my experiments, make adjustments as needed, and better understand the experimental processes.
Actor:	Student
Goal:	The student wants to have access to real-time information about the progress and status of the experiments they are currently conducting in the virtual lab.
Pre-conditions:	● The student has access to the Dexter's Laboratory: A VR Laboratory Simulator application. ● The student must have started a lab session, through the experiments menu ● The student is currently performing one or more experiments in the virtual lab environment.
Main Flow:	1. The student launches the Dexter's Laboratory application and joins the virtual lab activity by selecting one of the experiments in the experiments menu or using the provided access code. 2. The student is transported to the virtual laboratory 3. Student performs task relevant to the selected experiment (see use case 2.1.3) 4. As the student is conducting an experiment, they can access the "Experiment Progress" interface, which may be displayed on a dedicated panel or accessed through a menu option.

2.1.9 View History of Performed Lab Activities and Grades/Scores

Description	As a Student, I want to be able to view the history of the lab activities I have performed in the virtual lab environment, along with the grades or scores I have received, so that I can track my progress, identify areas for improvement, and demonstrate my achievements.
Actor:	Student
Goal:	The student wants to have access to their historical record of completed lab activities and the associated grades or performance scores.
Pre-conditions:	• The student has access to the Dexter's Laboratory: A VR Laboratory Simulator application. • The student has completed one or more lab activities within the virtual lab environment.
Main Flow:	1. The student opens the Dexter's Laboratory app and navigates to the "Lab History" section, which they can access through a menu, dashboard, or dedicated interface 2. In the Lab History section, the student can view the following information: a. A chronological or categorized list of the lab activities the student has completed in the virtual environment b. Performance scores or grades: The grades or scores the student has received for each completed lab activity, based on factors such as: i. Accuracy of measurements and observations ii. Adherence to experimental procedures and safety protocols iii. Interpretation and analysis of results iv. Overall experiment success rate c. Dates of completion: The dates when the student completed each lab activity. 3. With the lab history information, the student can reflect on their learning progress and identify areas where they can improve.

2.1.10 Select Pre-Set Experiments for Self-Directed Learning in the Virtual Lab

Description	As a Student, I want to be able to select and perform pre-set experiments in the virtual lab environment, so that I can engage in self-directed learning and explore chemistry concepts at my own pace.
Actor:	Student
Goal:	The student wants to have the ability to select and conduct pre-set experiments in the virtual lab to support their self-directed learning.
Pre-conditions:	• The student has access to the Dexter's Laboratory: A VR Laboratory Simulator application.
Main Flow:	1. The student opens the Dexter's Laboratory app and navigates to the "Experiment Selection" section or interface. 2. In the Experiment Selection area, the student can view a catalog or menu of the available pre-set experiments. 3. The student can browse the experiment options, which may be organized by topic, difficulty level, or specific chemistry concepts. 4. Based on their interests, learning goals, or current understanding of the subject matter, the student selects the pre-set experiment they would like to perform. 5. The student then launches the selected experiment in the virtual lab environment and begins the self-directed learning process.
Alternative Flow	If the student wants to access additional information or guidance related to the pre-set experiment, they can navigate to the tutorial within the virtual lab interface.

2.2 Functional Requirements

2.2.1 Module 1: Asset Creation

Following are the requirements for module 1:

1. FR1: The system shall use interactive 3D modeling for a fully immersive environment.
2. FR2: The system shall fully support 3D equipment creation with precise textures and color schemes.
3. FR3: The system shall provide smooth player movement to allow easy navigation for the students in the VR environment.

2.2.2 Module 2 : Mechanic Design

Following are the requirements for module 2:

1. FR1: The system shall use a real-time engine to stimulate the chemical reactions such as the mixing of acid and bases.
2. FR2: The system shall provide laboratory apparatus handling allowing students to work on lab tools like burettes, pipettes, and thermometers etc.
3. FR3: The system shall support pouring of liquid and measure mechanics, guaranteeing precise volume calculations for research.
4. FR4: The system shall manage sample tutorials and guidance walking students through different chemistry experiments.

2.2.3 Module 3: Level Design

Following are the requirements for module 3:

1. FR1: The system shall cover educational experiments on acids, basis and salts reactions for the understanding of 9-10 and O levels students.
2. FR2: The system shall teach students of 9-10 and O levels about acidity and alkalinity by enabling pH testing using multiple indicators.
3. FR3: The system shall teach students of 9-10 and O levels about conduction of cation and anion tests for substance analysis.
4. FR4: The system shall demonstrate heat exchange experiments by simulating exothermic and endothermic reactions in the VR environment for enhanced understanding of the students.
5. FR5: The system shall conduct experiments for the better understanding of the students regarding the physical states of matter (solid, liquid and gas).
6. FR6: The system shall perform experiments on chemical reactivity based on the set curriculum of 9, 10 and O levels for better understanding of the students.
7. FR7: The system shall walk through the basic organic chemistry concepts making it easier for the students to understand and explore different kinds of chemical reactions and compounds.

2.2.4 Module 4: Application Design

Following are the requirements for module 4:

1. FR1: The system must offer a virtual safety protocol that includes virtual safety equipment and hazard warnings to make sure students follow correct safety

measures.

2. FR2: The system shall have a user-friendly interactive UI that displays real-time feedback on the experiment performance and instructions for the experiment.
3. FR3: The system shall walk the users through tutorial and guidance to ensure better understanding on how to interact with the virtual equipment, environment and tools, thus, maintaining a tutorial and guidance system.
4. FR4: The main menu interface of the system will allow users to choose and launch various experiments.

2.2.5 Module 5: Testing

Following are the requirements for module 5:

1. FR1: The system must verify that instruments and equipment are operating as intended, in order to guarantee precise and quantifiable mechanics throughout experiments.
2. FR2: The system will be tested to make sure it runs well on all platforms, with consistent frame rates and few performance bottlenecks.
3. FR3: The system shall guarantee smooth movement of a player and accurate collision detection in the VR based environment.
4. FR4: To ensure the experiment's sequence is correct, the system shall perform level-based testing.

2.2.6 Module 6: Deployment and Optimization

Following are the requirements for module 5:

1. FR1: The system shall undertake performance optimization to guarantee efficient functionality on the Meta Quest platform.
2. FR2: The system will have improved visuals and graphics for immersive experience and realistic stimulation.
3. FR3: The system will smoothly integrate with Meta Quest Platform using Meta Quest SDK.
4. FR4: The system will go through final UX/UX tweaks to enhance overall user experience and user engagement.

2.3 Non-Functional Requirements

2.3.1 Reliability

1. REL-1: The application shall ensure that it stores the user's progress and saves the main phases of an experiment after their completion.
2. REL-2: The system should be able to replicate the reactions of chemical compounds of each experiment carried out as close to the real world as possible with a maximum difference of 2%.
3. REL-3: The system should be able to handle a minimum of 500 users without experiencing any delay in the execution or inaccuracy of the simulations that are generated.

2.3.2 Usability

1. USE-1: Users should be able to begin a new experiment from the home screen with no more than five clicks or interactions from the home menu.
2. USE-2: All new users will be able to complete the basic tutorial and a small experiment conducted within the first fifteen minutes of starting the virtual lab.
3. USE-3: The application should provide the users with a facility that allows them to switch between controller and hands by using the settings icon.

2.3.3 Performance

1. PER-1: The system should be able to respond to the user's interaction with the environment within 50 milliseconds to provide a real-time experience.
2. PER-2: Experiments conducted by the user shall be calculated and displayed in real-time without any noticeable delay from the system.

2.3.4 Security

1. SEC-1: The application shall have access to admin control that allows the admin to fully manage the experiments such as adding, deleting, or modifying.
2. SEC-2: The system shall not save any other personal information of the user apart from basic requirements of the system and progress tracking.

2.3.5 Compatibility

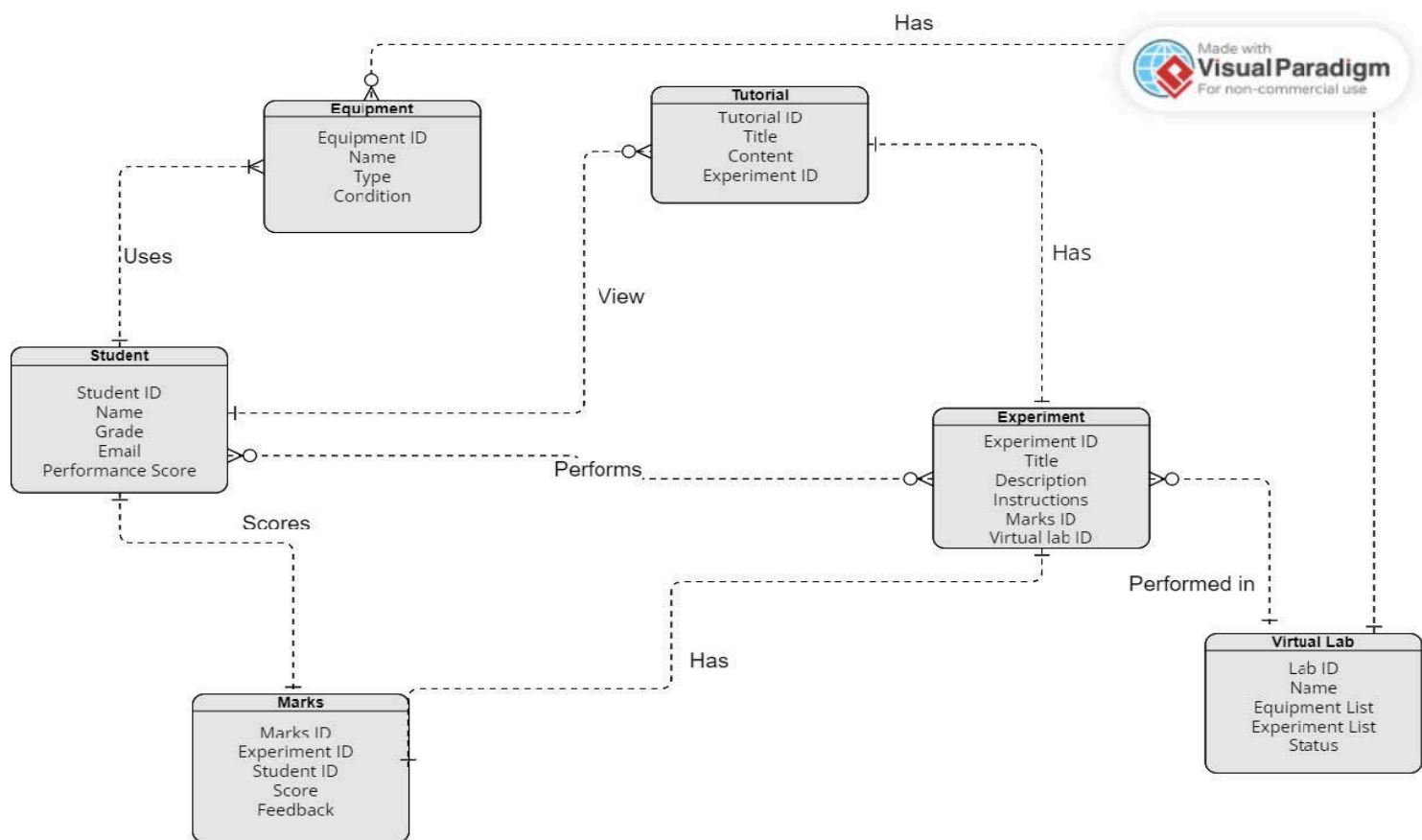
1. COM-1: The system must be fully compatible with Meta Quest 2, and utilize the VR-based functionality across multiple platforms.

2.3.6 Scalability

1. SCA-1: The application must be capable of storing and managing data for at least 500 individual users and retrieving the data without compromising performance.
2. SCA-2: The system must be able to accommodate future expansions, allowing new experiments and modules to be added without any restructuring of the application.

2.4 Domain Model

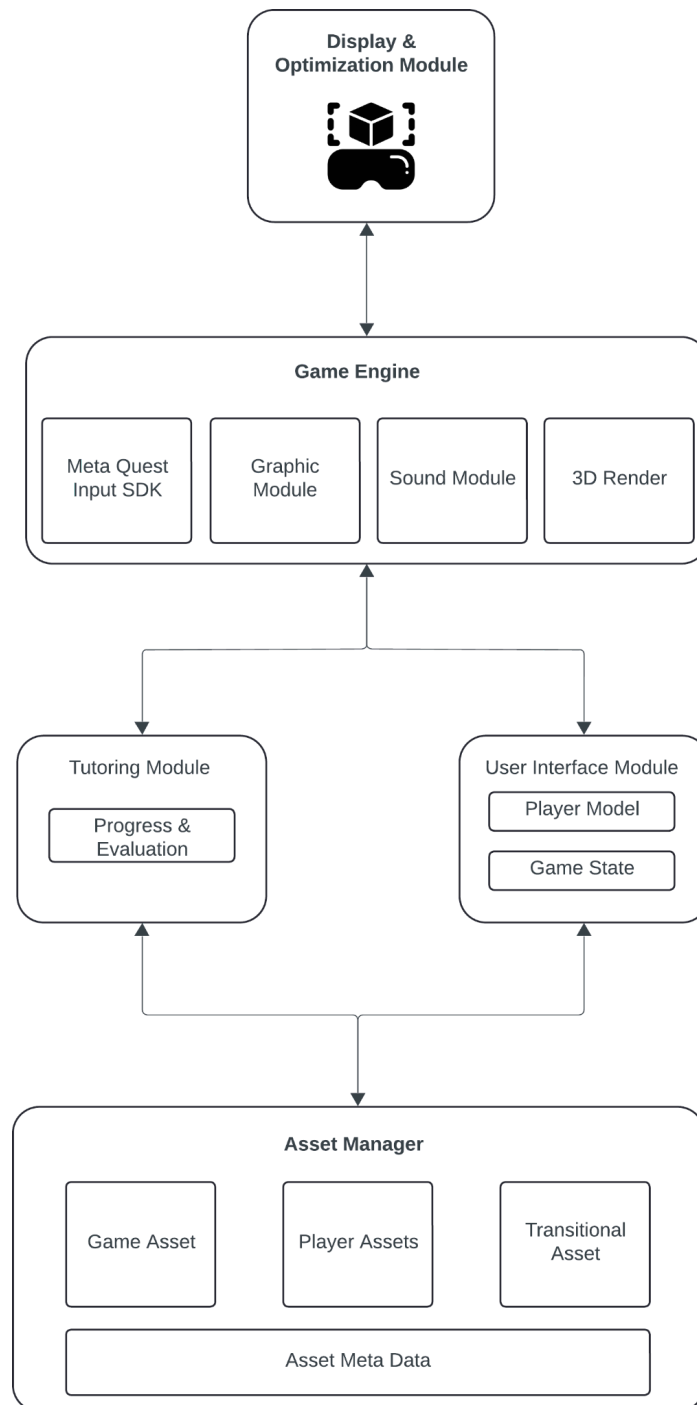
Create a representation of the domain model for your project.



2.4 :Domain Model

Chapter 3

3.1 Architectural Design



3.1 : Architecture Diagram

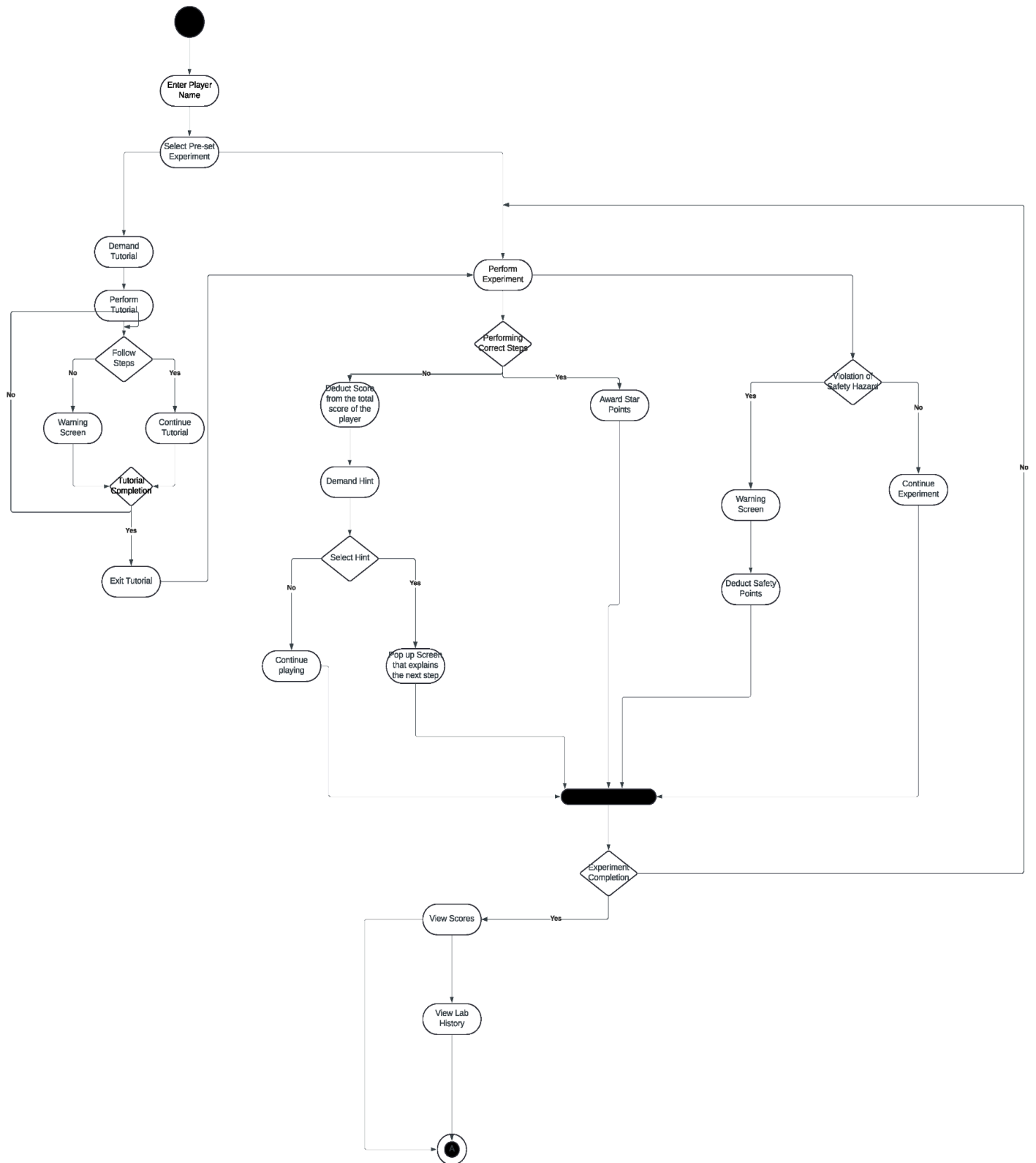
3.2 Design Models

Design Models for Procedural Approach

“The applicable models for the project using procedural approach may include:

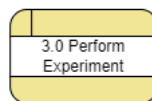
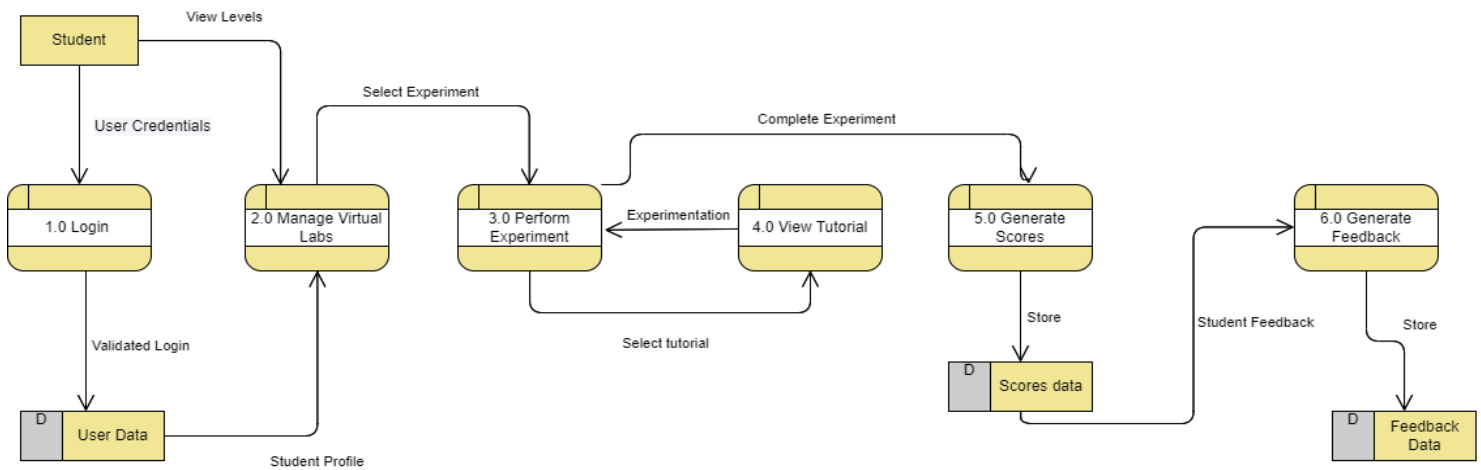
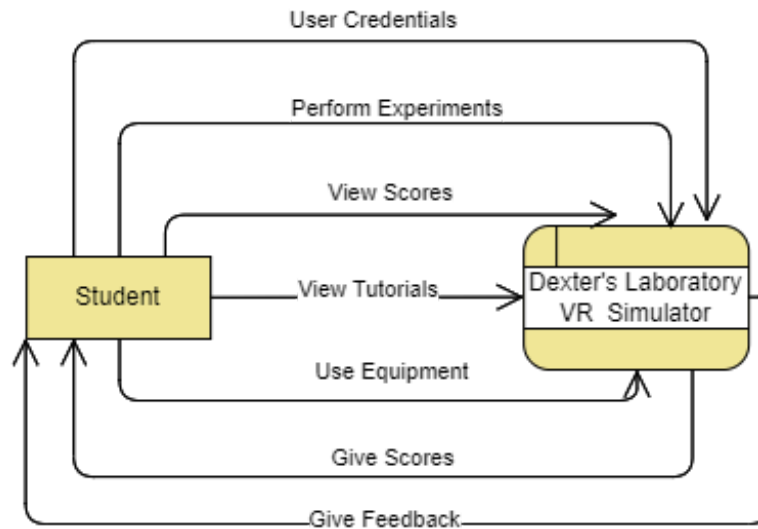
- Activity Diagram
- Data Flow Diagram (data flow diagram should be extended to 2-3 levels. It should clearly list all processes, their sources/sinks and data stores.)
- System-level Sequence Diagram
- State Transition Diagram (for the projects which include event handling and backend processes)”

3.2.1 Activity Diagram



3.2.1 : Activity Diagram

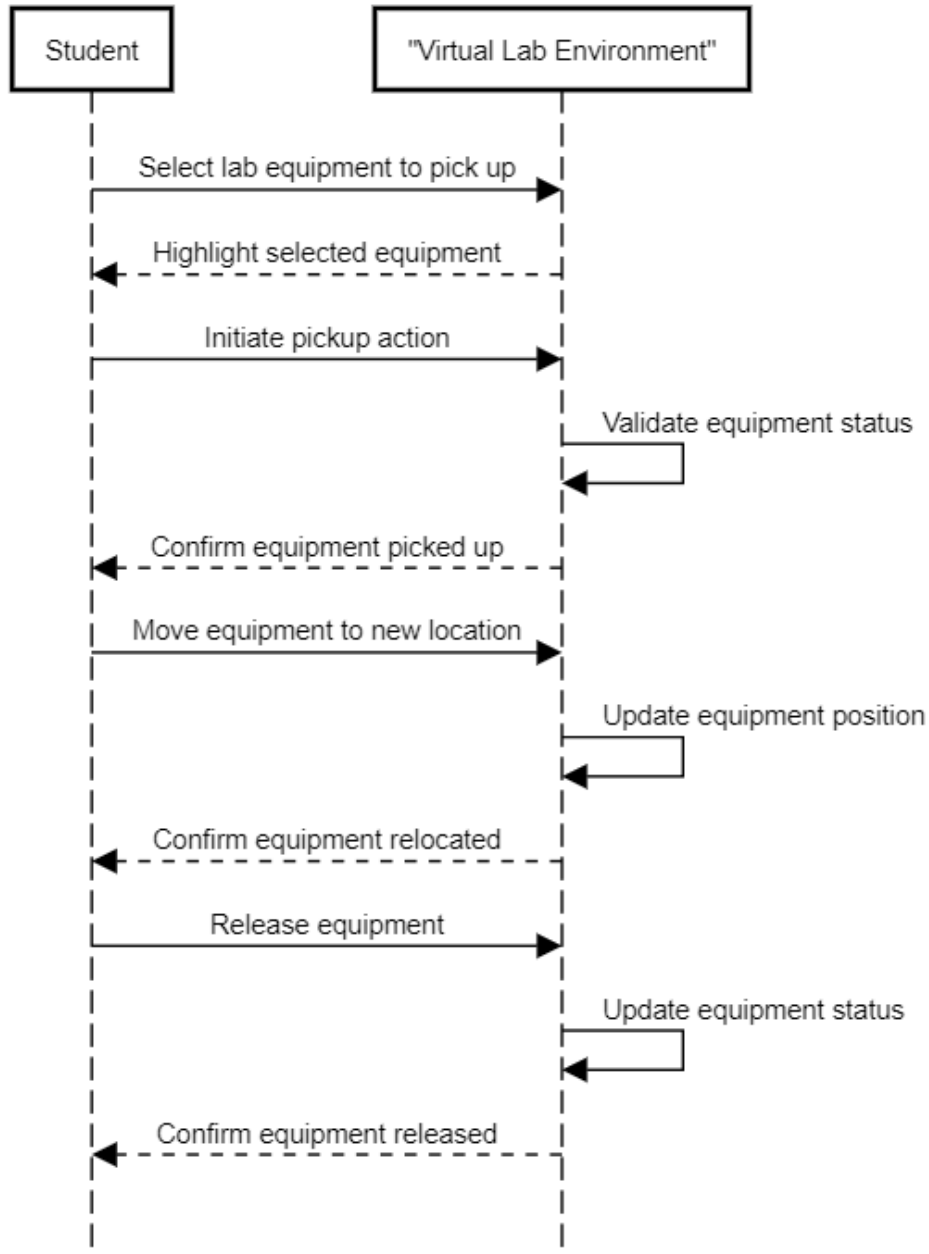
3.2.2 Data Flow Diagram



3.2.2 : Data Flow Diagram Level 1,2 and 3

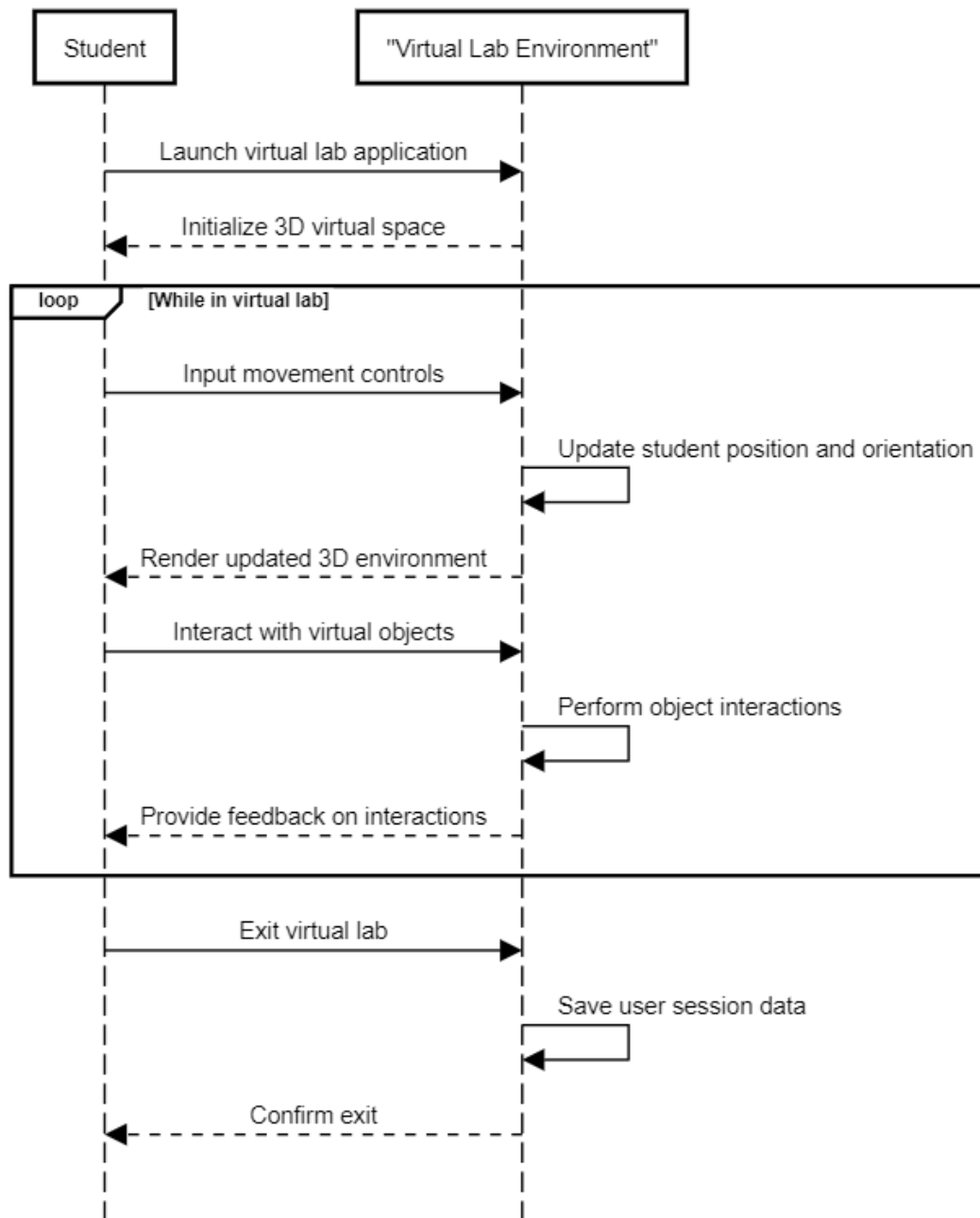
3.2.3 System Level Sequence Diagram

3.2.3.1: Pickup and relocate (pipettes, burettes, thermometers, etc.)



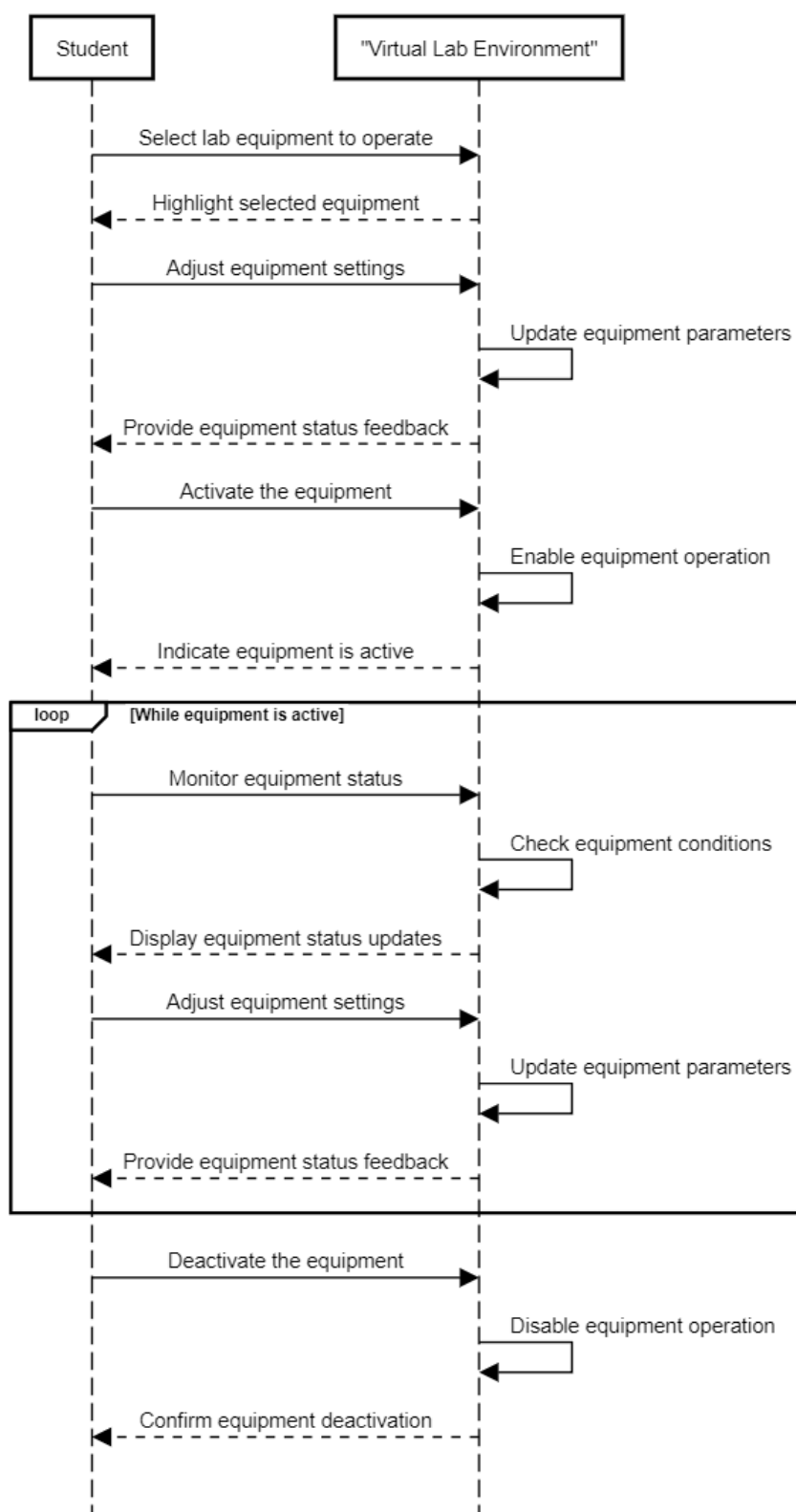
3.2.3.1: Pick up and relocation

3.2.3.2: Move around in the 3d space of the virtual laboratory



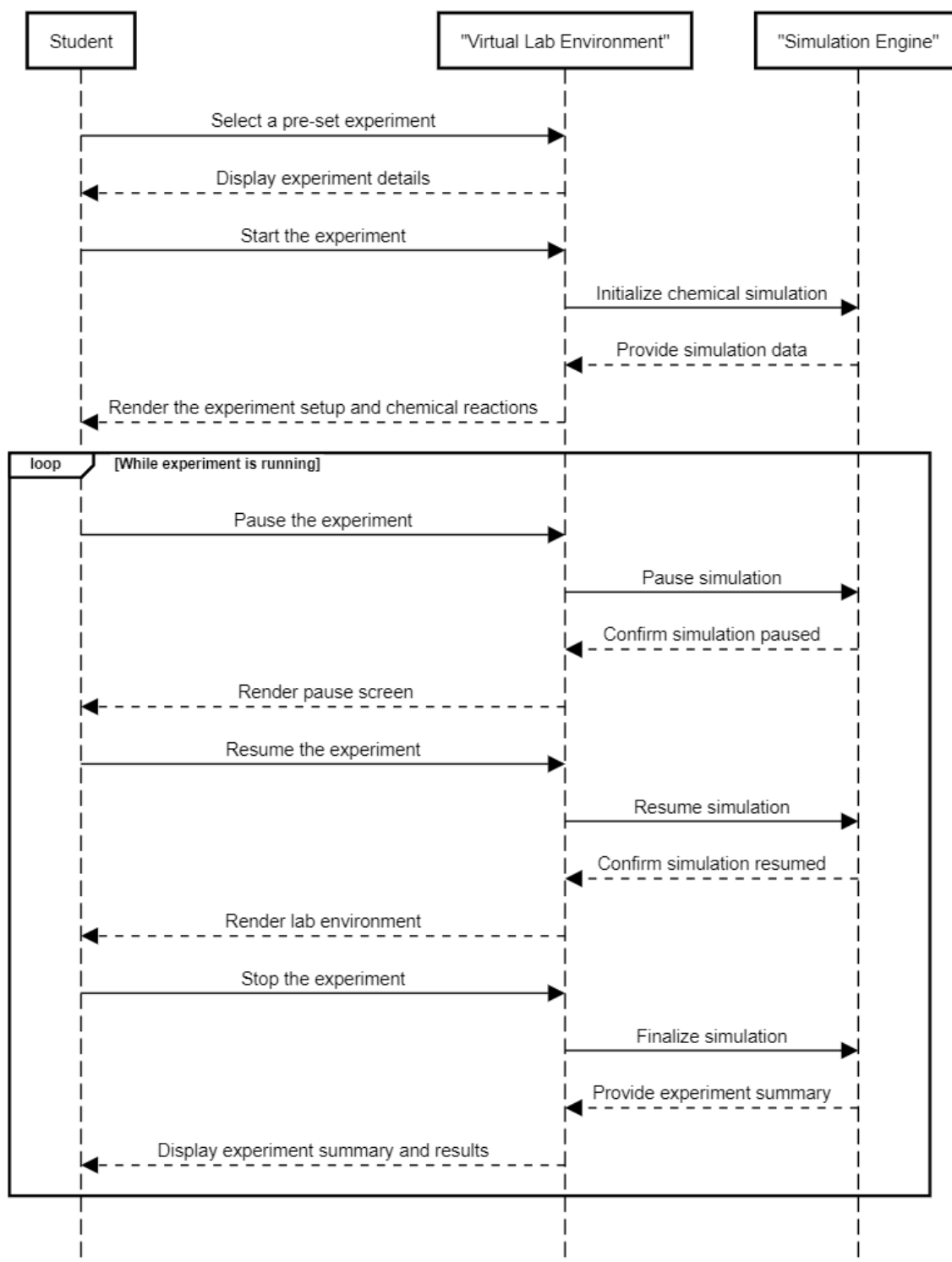
3.2.3.2: Moving around in 3d space of Virtual Reality

3.2.3.3: Observe and learn from the simulated chemical reactions and processes



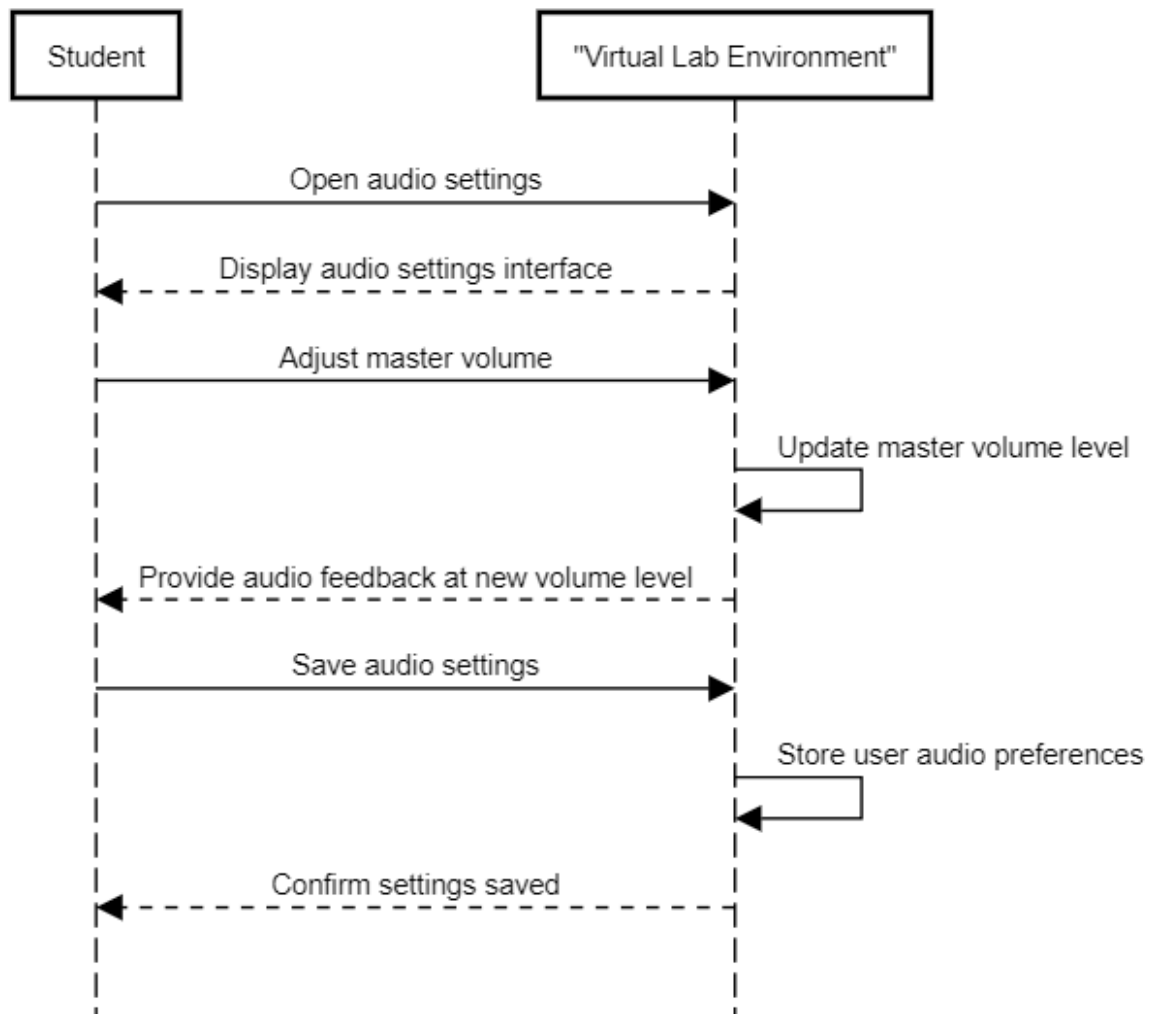
3.2.3.3: Observe and learn from the simulated chemical reactions and processes

3.2.3.4: Pause, start, and stop experiments



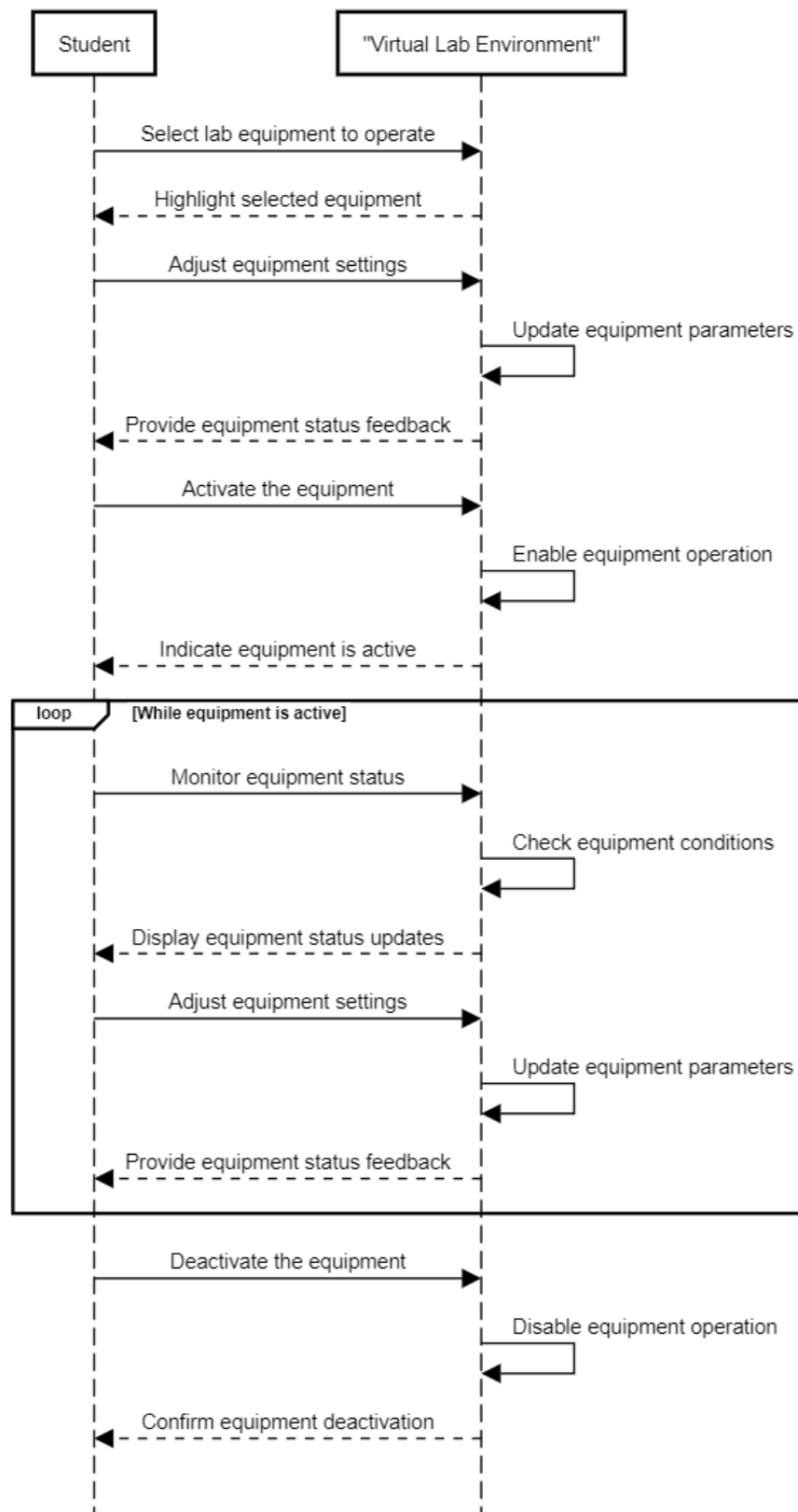
3.2.3.4: Pause, start and stop reactions

3.2.3.5: Adjust sound levels



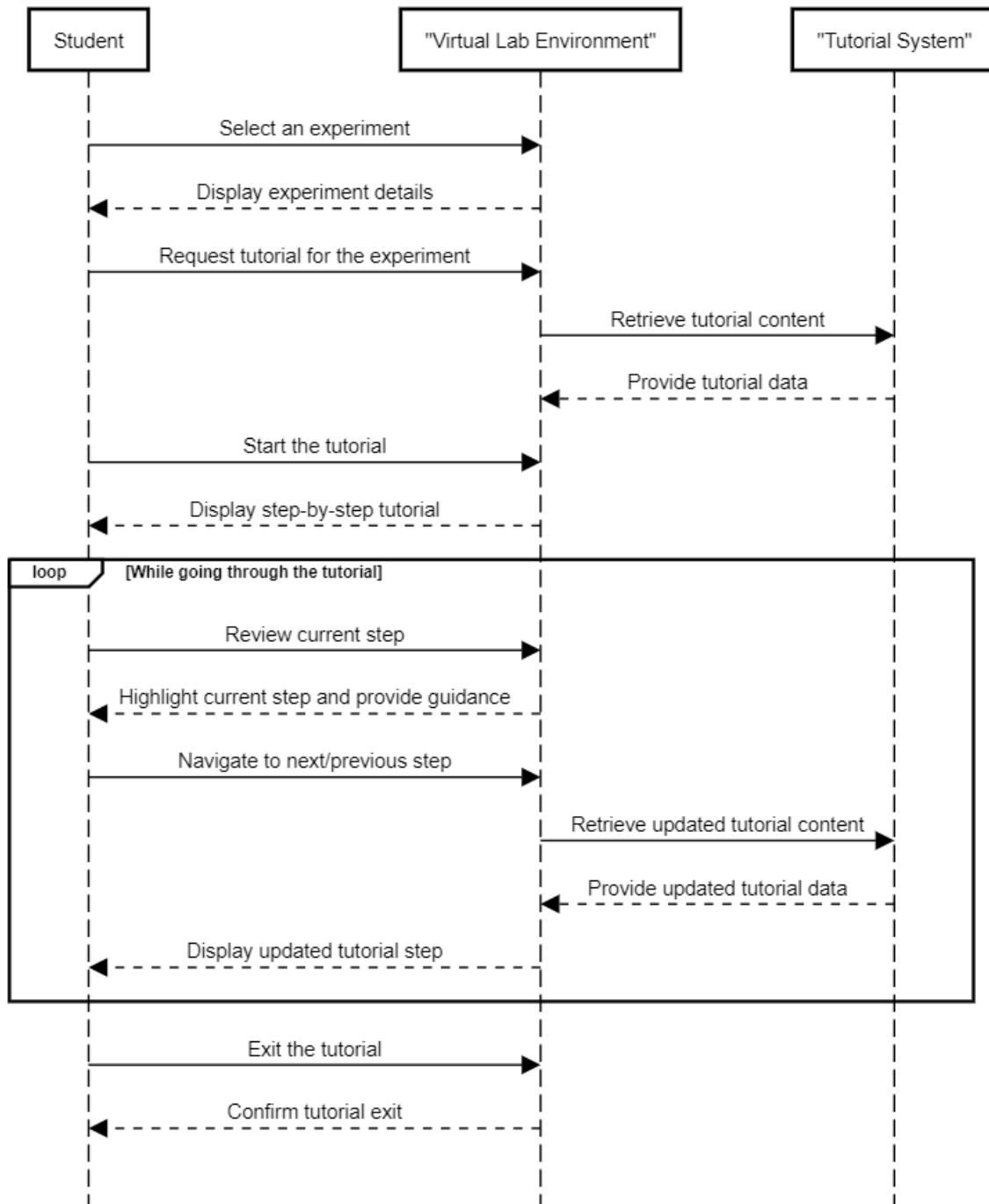
3.2.3.5: Adjust Sound levels

3.2.3.6: Operate Bunsen burners and other equipment



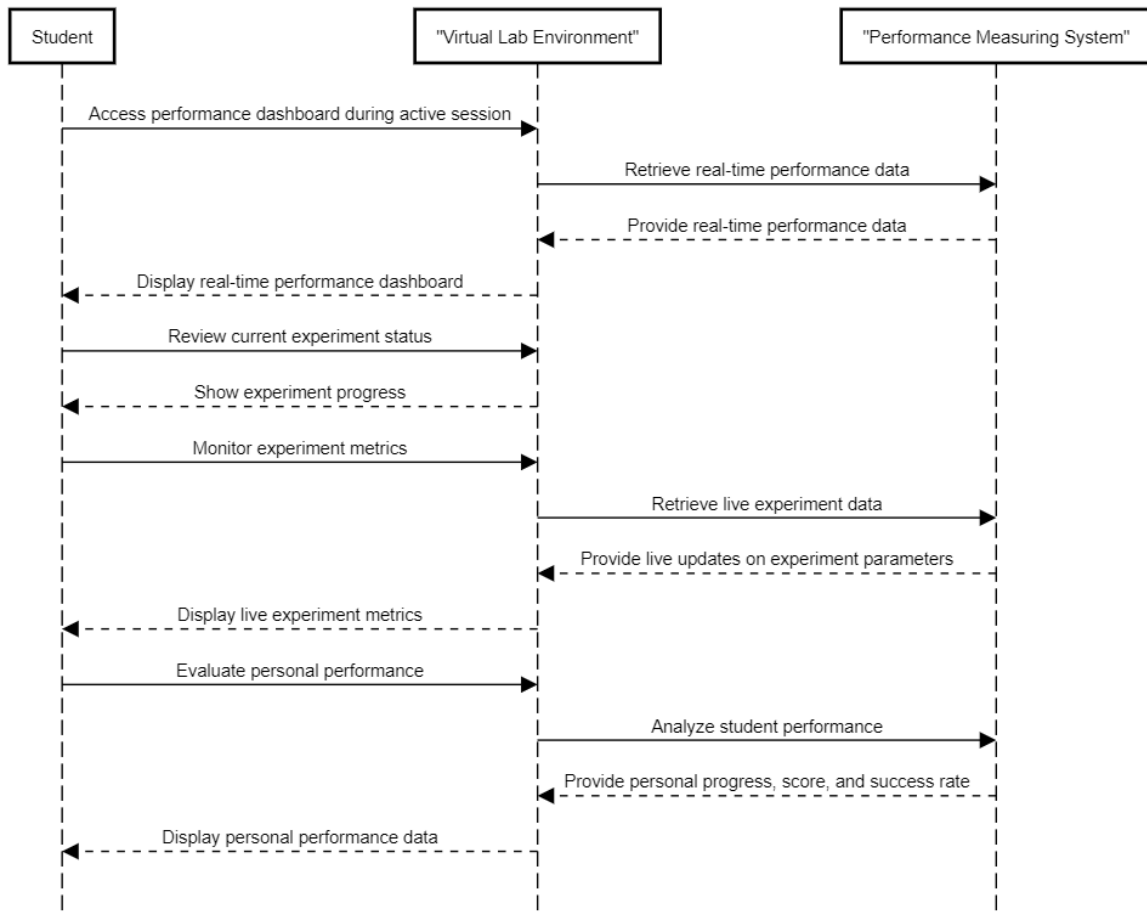
3.2.3.6: Operate Bunsen burners and other equipment

3.2.3.7: Access step-by-step tutorials and guidance for experiments



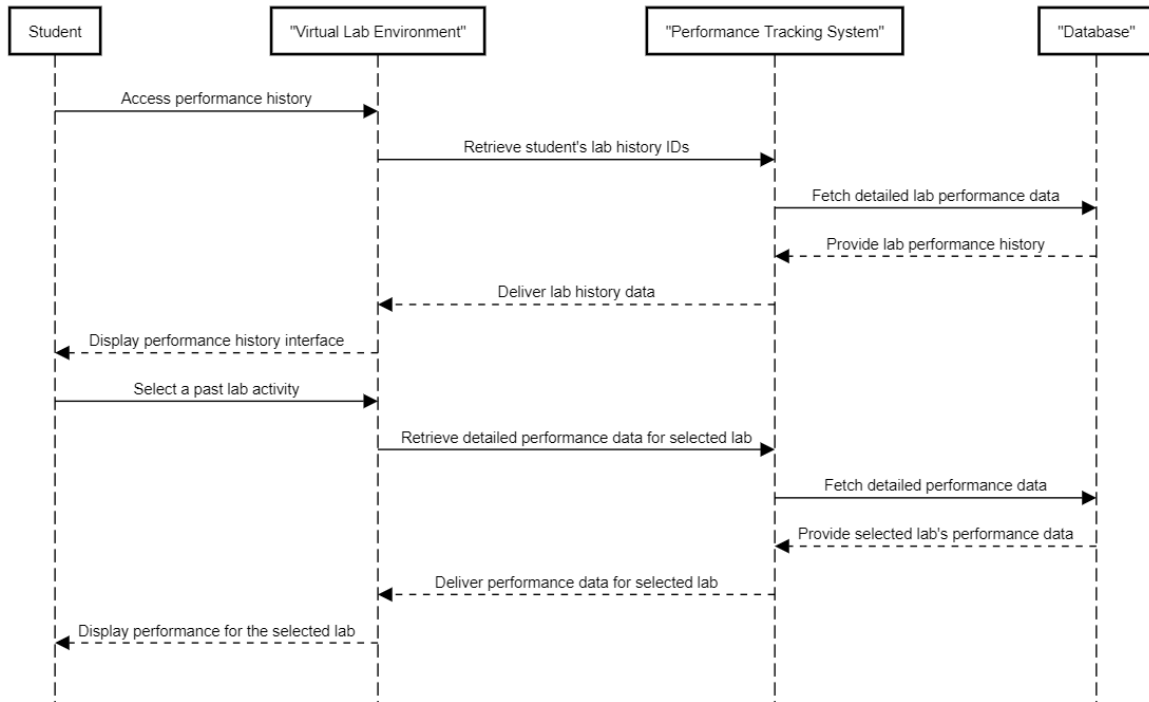
3.2.3.7: Access step-by-step tutorials and guidance for experiments

3.2.3.8: View progress and performance in the virtual lab activities



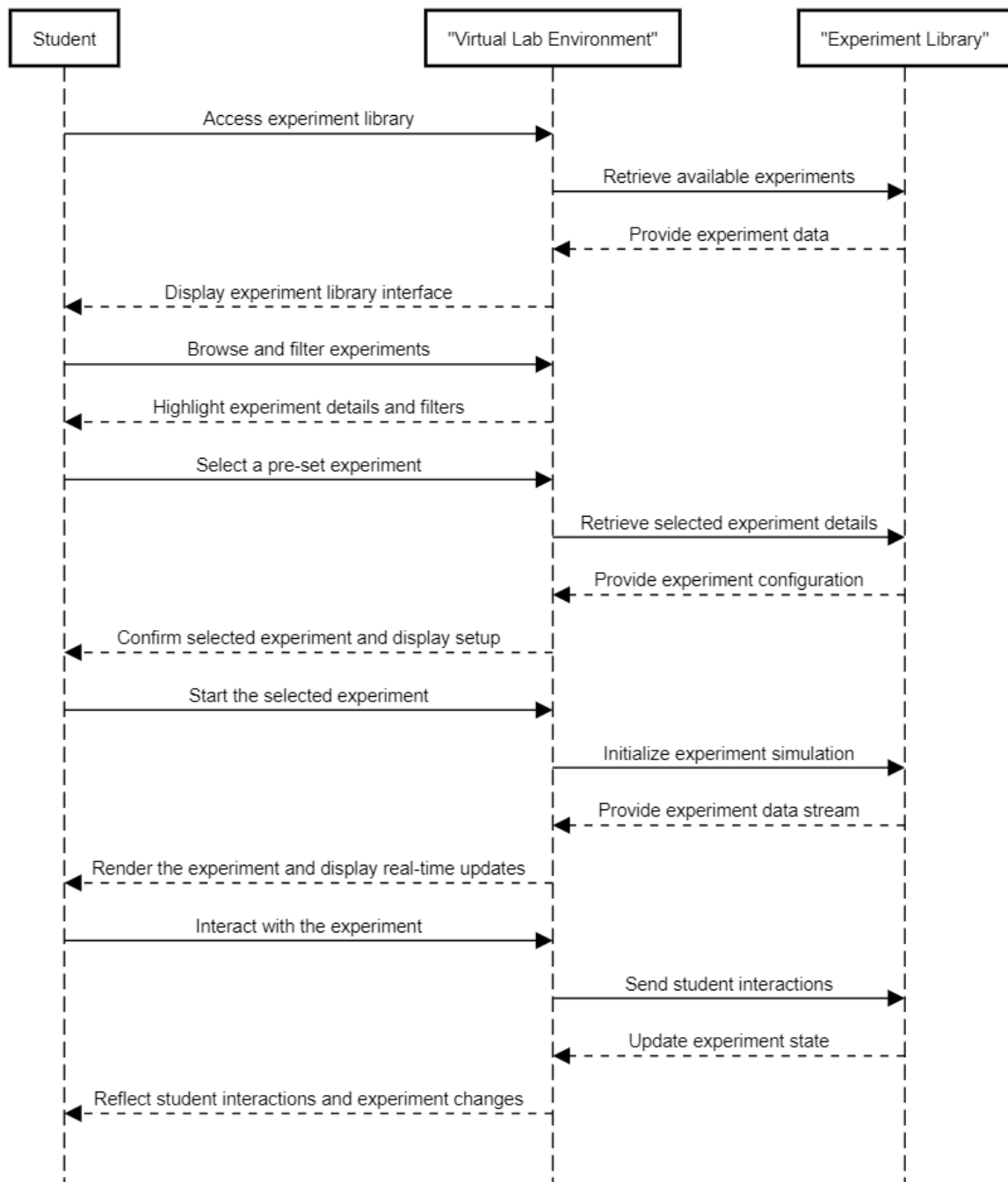
3.2.3.8: View progress and performance in the virtual lab activities

3.2.3.9: View History of performed lab and their grades / scores



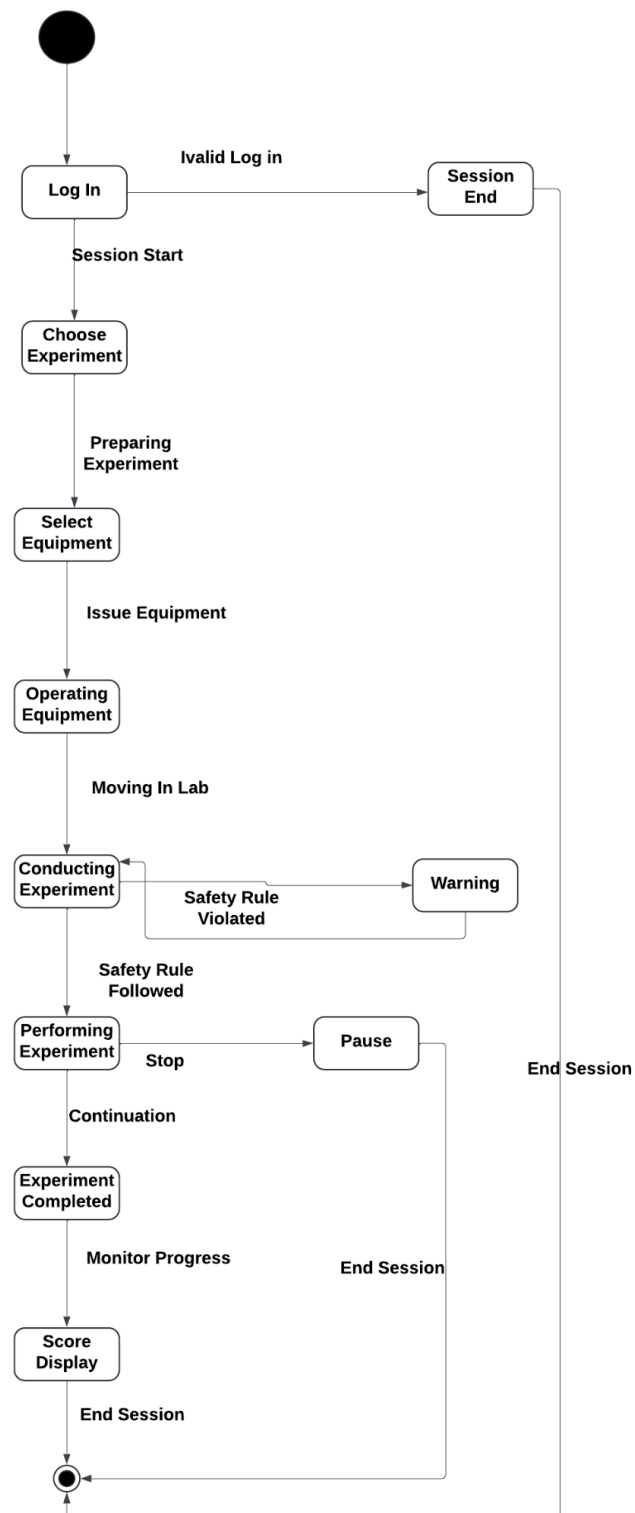
3.2.3.9: View History of performed lab and their grades / scores

3.2.3.10: Select pre-set experiments for self-directed learning



3.2.3.10: Select pre-set experiments for self-directed learning

3.2.4 State Transition Diagram

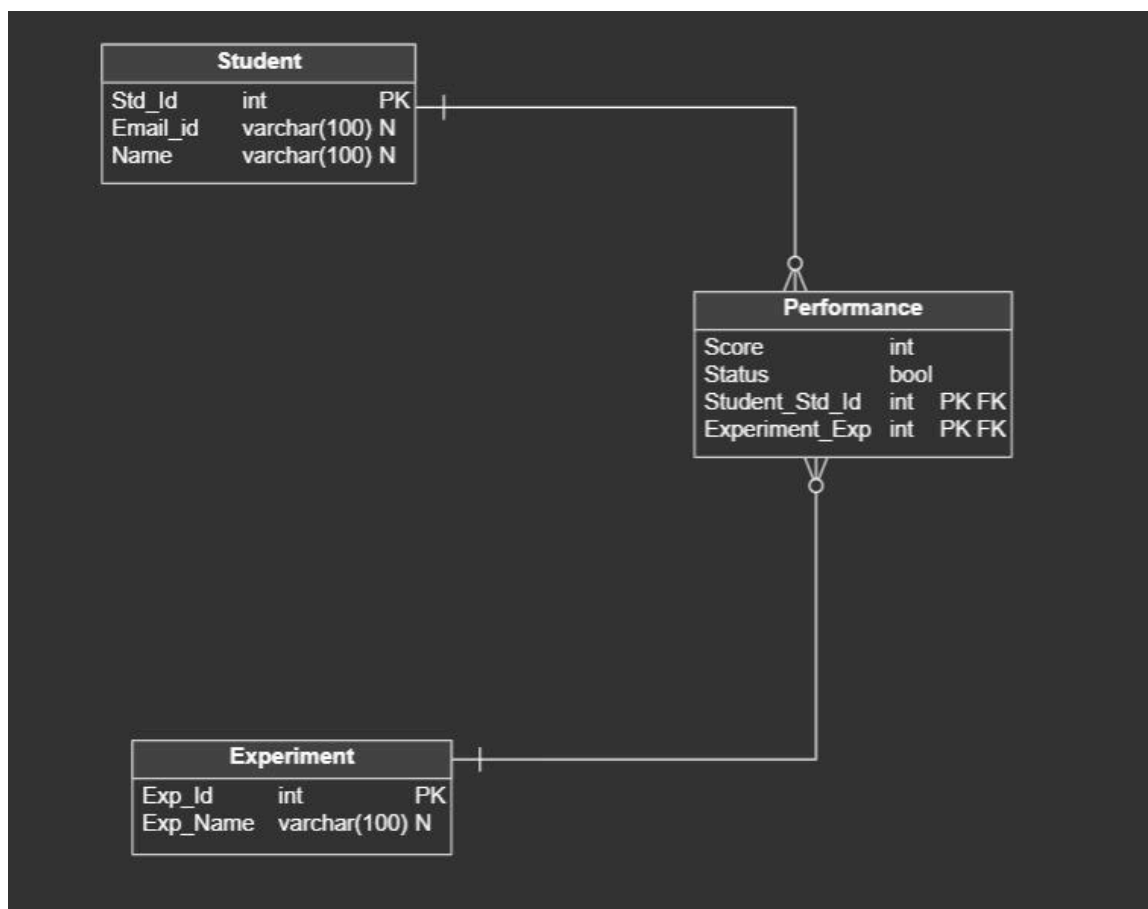


3.2.4 :State Transition Diagram

3.3 Data Design

The data design for this virtual lab application should focus on maintaining the session status for each player. To achieve this, the student's credentials and all their progress throughout the experience will be stored in the backend. This includes the scores they have achieved and the completion status of the experiment levels they have successfully completed.

By storing this data, the system will be able to retrieve the student's history whenever they log in to their account. Given the relatively simple nature of the data, the team would likely find that a non-relational database, such as MongoDB, would be a preferred choice for storage and management. This NoSQL database solution can efficiently handle the storage and retrieval of the student-related data without the need for a complex relational database structure, providing a streamlined and scalable data management system for the virtual lab application.



3.3 : Data Design

Bibliography

- [1] Virtual reality in an unreal engine. <https://www.unrealengine.com/en-US/xr>. Accessed: 2024-10-06.
- [2] Meta quest development. <https://developer.oculus.com/quest/>. Accessed: 2024-10-06.
- [3] Unreal engine 5.3.2 for meta quest vr | community tutorial. <https://dev.epicgames.com/community/learning/tutorials/3Vx6/unreal-engine-5-3-2-for-meta-quest-vr>. Accessed: 2024-10-06.
- [4] S. Jain, "Activity diagrams | unified modeling language (uml)," <https://www.geeksforgeeks.org/unified-modeling-language-uml-activity-diagrams/>. Accessed: 2024-10-06.
- [5] S. Jain, "What is dfd(data flow diagram)?," <https://www.geeksforgeeks.org/what-is-dfddata-flow-diagram/>. Accessed: 2024-10-06.